Immune changes in tumor tissue & lymph nodes associated with aggressive non-small cell lung cancer (NSCLC)

Vaidehi Gupta
PI: Dr. Josh Campbell, Dr. Sarah Mazzilli
Campbell Lab (Boston University School of Medicine)
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THE PREPRINT SERVER FOR HEALTH SCIENCES

Multimodal single-cell and spatial profiling reveals altered T cell-mediated immunity and B-cell follicular architecture in non-metastatic lymph nodes of patients with aggressive non-small cell lung cancer

Zhan Hao Xi, Yusuke Koga, Shannon McDermott, Erin E. Kane, Roxanna Pfefferkorn, Ehab Billatos, Paul R. Hosking, Jennifer E. Beane, Eric J. Burks, Sarah A. Mazzilli, Kei Suzuki, Joshua D. Campbell

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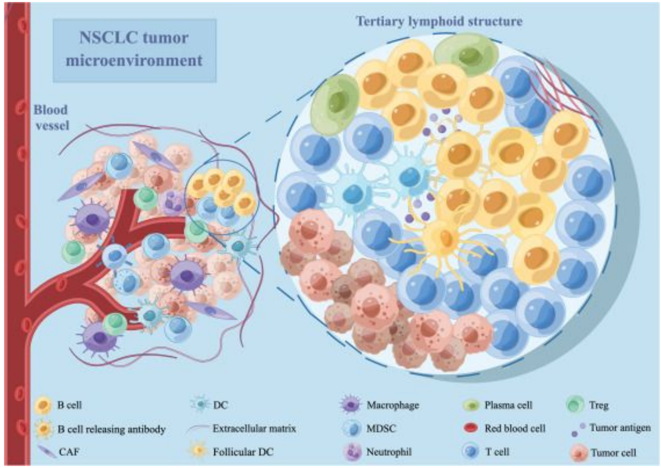
Why does NSCLC matter?

- **Non-small lung cancer (NSCLC)** accounts for ~85% of lung cancer cases.
- Advanced NSCLC has poor survival outcomes despite modern therapies.
- Tumors can evade the immune system by reshaping the tumor microenvironment and nearby lymph nodes.



Understanding how immune cells change during aggressive disease can improve…

- Prognosis
- Immunotherapy strategies

The Study

Collected 36 lymph node tissues from 11 NSCLC patients from Boston Medical Center.

Experimental Approach

- Multi-modal spatial and single-cell profiling of lymph nodes
  - Single-cell RNA sequencing
  - CITE-seq
  - Imaging Mass Cytometry (IMC)

What they did

- Identified cell populations using single-cell clustering
- Characterized spatial organization of immune cells in lymph nodes
- Compared immune architecture across disease states

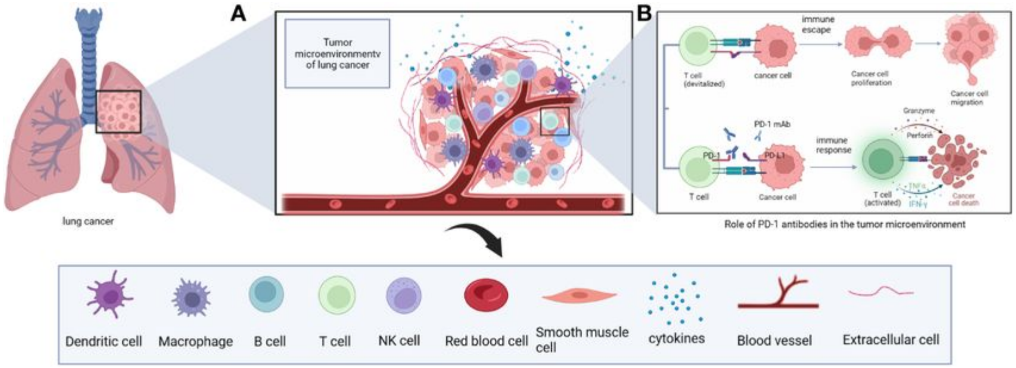
Overall: They found coordinated immune dysregulation in aggressive NSCLC.

The gap in knowledge

- **What's known:** Lymph node metastasis is already used to clinically stage NSCLC.
- The pre-print characterized immune dysregulation in regional lymph nodes of NSCLC patients
- **Less is known about how the tumor immune microenvironment itself changes across disease stages.**
- Also unclear which immune features are…
  - Shared between tumors and lymph nodes
  - Or are they unique to each tissue compartment

Research Question

How does the tumor immune microenvironment in NSCLC change across disease stages?



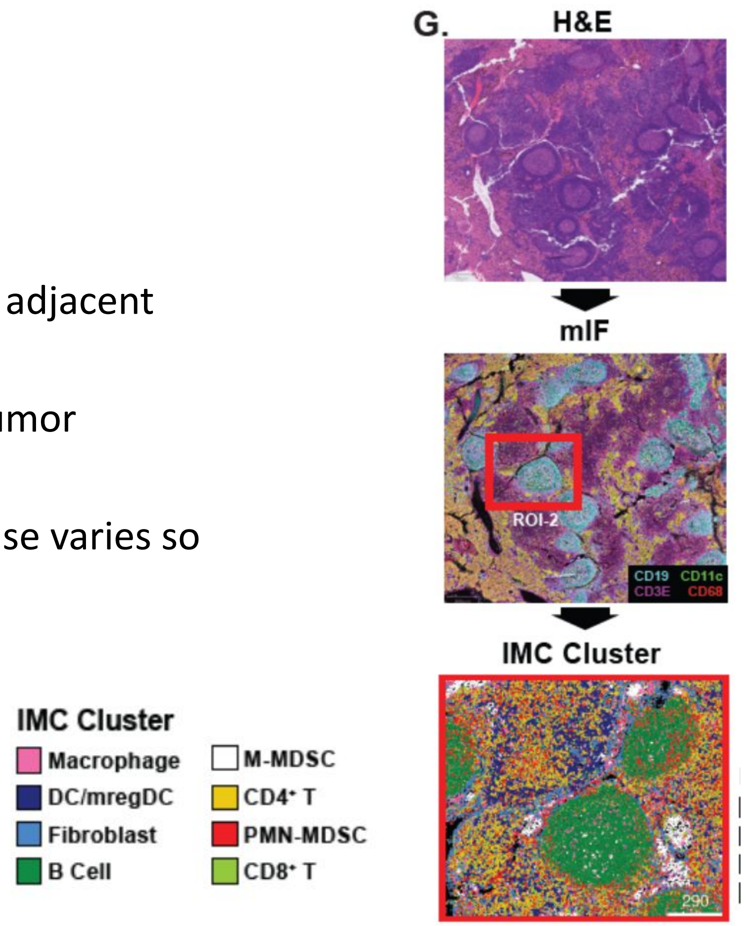
High-level Goals

1. **Characterize the tumor immune microenvironment**
   - Identify immune cell populations + spatial niches in tumor and adjacent normal tissue
2. **Compare across disease stages**
   - Determine how tissue organization changes in aggressive disease
3. **Correlate with lymph node findings**
   - Identify shared vs. compartment-specific immunosuppressive features

The Data

Imaging Mass Cytometry Data

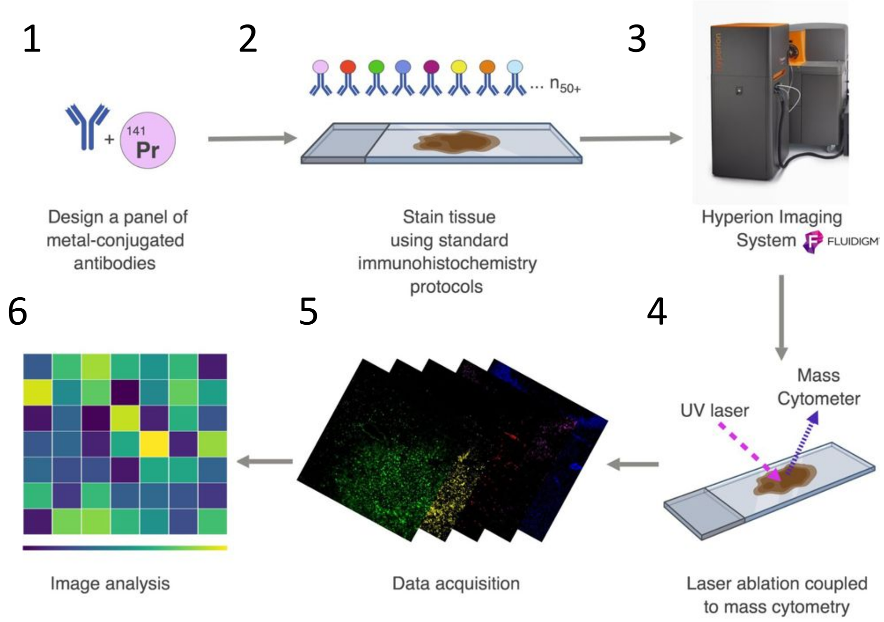
- Generated from multiple regions of tumor and adjacent normal tissue from each patient.
- Analyzing this data can help characterize the tumor microenvironment of NSCLC.
- Allows us to understand why treatment response varies so much across patients



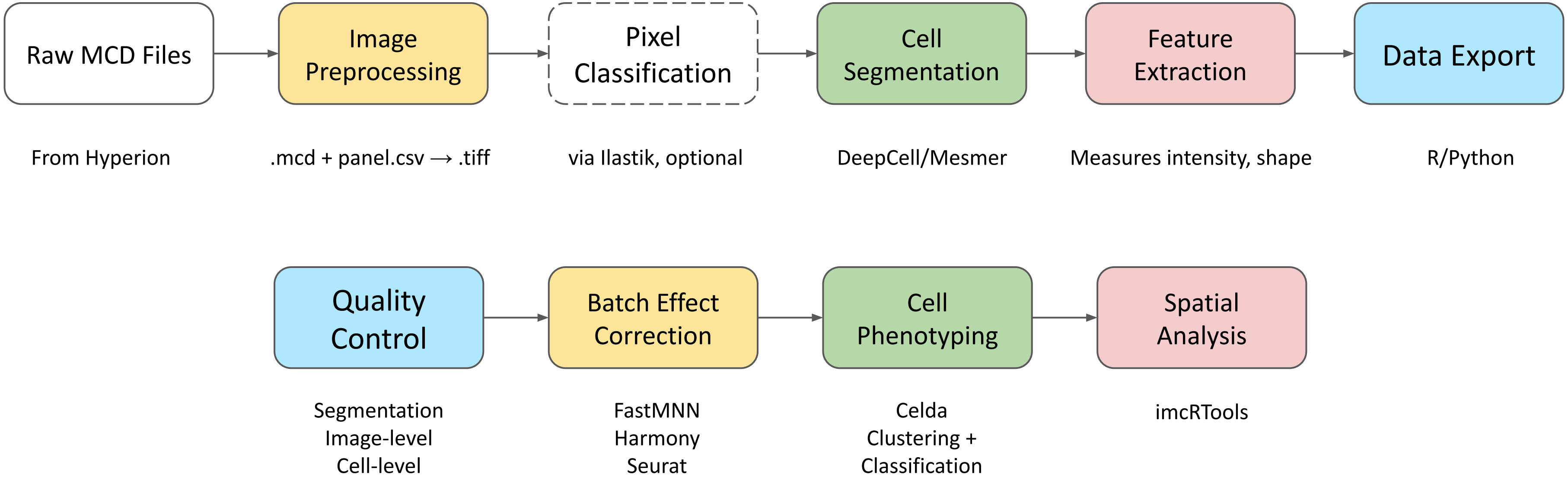
What is Imaging Mass Cytometry (IMC)?

A spatial proteomics technique that can detect 40+ protein markers at single-cell resolution in tissue samples.

- Used to perform highly multiplexed imaging.
- Combines metal-labeled antibodies + laser ablation mass spectrometry



IMC Data Analysis Workflow



Timeline

<u>**Week 1-2**</u>

- **Understand the pre-print and how IMC works**
- **Visualize the raw data using MCDViewer (on a subset of ROIs)**
  - **Make histograms + density plots**
  - **Look for artifacts in the data**

<u>Week 3-5</u>

- Run Steinbock pipeline on tumor samples
  - Preprocessing → segmentation → feature extraction
- QC report (segmentation quality, image-level, cell-level)
- Import data into R using imcRtools

<u>Week 6-7</u>

- Batch correction if needed
- Cell phenotyping + cell type composition analysis ← ***This is where we start splitting up the work***

<u>Week 8-10</u>

- Run spatial niche analysis
- Correlation with lymph node IMC findings from the pre-print
- Write project reports and prepare posters for presentation

Thank you!
Any questions?